Non-contextual meanings and funtional readings of *wh*-questions

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1 Introduction

Three readings of a *wh*-question with a quantifier:

(1) Q: Who does everyone love?
A: Sue (individual reading)
A′: Their mother. (functional reading)
A″: John loves Sue; Bill loves Mary; Tim loves Amy. (list reading)

Proposal by many linguists like Groenendijk and Stokhof (1984), Engdahl (1980, 1986), and Chierchia (1993), Hornstein (1995), Sharvit (1997,1999), Dayal (1996, 2016), Xiang (2023), etc.:
A *wh*-phrase with a functional reading carries two or more indexes, one of which is bound by a *wh*-phrase and the other(s) by (a) quantifier(s).

(2) Who$_i$ does everyone$_j$ love t$_i^j$?
[who f does [everyone$_x$ love $f(x)$]]

4 Problems

- In natural language syntax, it has been assumed that a movement of a phrase leaves one trace with one index.
- If the *wh*-phrase denotes a function, it should be *what*, not *who*.
- If there is more than one quantifier in a sentence, the trace of a *wh*-phrase may need three or more indexes.

(3) a. Q: To whom does every writer dedicate a book?
A: To the person that inspired her/him to write it.

b. [[to whom]$_i$ [every writer]$_j$ [a book]$_k$ [t_j dedicates t_k $t_i^{j,k}$]]

c. $\lambda p \exists f[p = \forall x[writer(x) \rightarrow \exists y[book(y) \wedge dedicate(x, y, f(x, y))]]]$

- In the short answer *Their mother* in (1), the pronoun is free.

(4) $[\![\text{Their mother}]\!] = \iota x[mother(x, y)]$
Cf. Required interpretation: $\lambda y \iota x[mother(x, y)]$

Purpose of this paper

To propose an analysis that assumes a single-indexed trace and that lacks the problems

2 Necessity of non-contextual meanings

2.1 Problems with binding relations

Reconstruction required:
In *wh*-questions, if a preposed phrase includes a DP that is subject to Binding Conditions (BCs), it must be reconstructed into the positions of its trace.

(5) a. ∗Which of *each other*'s friends did she remind t that *they* saw Bill?

b. Which of *each other*'s friends did *they* remind t that she saw Bill?

c. Which of *each other*'s friends did she remind Bill that *they* saw t ?

(6) ∗Whose examination of *John* did *he* fear?
⇒ ∗did *he* fear whose examination of *John* ?

It is reasonable to assume that **all preposed *wh*-phrases should be reconstructed**.

Intervention effects:
A *wh*-phrase cannot occur after a focused phrase.

(7) a. ??When did [only John$_F$] visit which museum?
b. Which museum did [only John$_F$] visit?

(8) Which of *each other*'s friends did she remind [only Bill$_F$] that *they* saw t?

⇒ We must get the reconstruction effect without actually putting a *wh*-phrase back.
It is reasonable to assume that **all preposed *wh*-phrase are not reconstructed but get the effect of reconstruction somehow.**

2.2 Non-contextual meanings

Intervention effects + Reconstruction effects $\Rightarrow$

- A preposed *wh*-phrase must be interpreted so that it has the scope of its trace.

  Question 1: How is a preposed *wh*-phrase interpreted?

- If a preposed *wh*-phrase has the scope of its trace, it is likely to behave like a *wh*-phrase in situ.

  Question 2: How is a *wh*-phrase in situ associated with the Mood?

Answer to Question 1: Non-contextual meanings

- An expression is usually interpreted w.r.t. a linguistic context.

- A linguistic context is $\langle g, w\rangle$, where g and w are an assignment and a possible world.

- The meaning of an expression with the feature [$+wh$] is a non-contextual meaning.

  non-contextual meaning of A: $\lambda\langle g, w\rangle[\![\alpha]\!]^{gw}$

- A *wh*-phrase and its trace carries the feature of $[+wh]$.
- A non-contextual meaning is contextualized with respect to a local context if the expression is in an IP.

(9)	$[_{MP}$	Wh-*phrase*$_i$	...	$[_{IP}$ ...	$t_i^{[+wh]}$	...	wh-*phrase*	...]]
		non-contextual			non-contextual		non-contextual	
		not contextualized			contextualized		contextualized	

- For the trace of a *wh*-phrase, we need a new assignment that assigns a non-contextual meaning.

  Contextual assignments: f, g, ...
  Non-contextual assignments: $\vec{f}$, $\vec{g}$, ...

(10) $[\![t_k^{[+wh]}]\!]^{\vec{g}g^*w^*} = \vec{g}(k)(\langle g^*, w^*\rangle)$

(5) c. $[_{MP}[_{DP}$Which of *each other*$_j$'s friends$]_k$ did she$_i$ remind Bill that *they*$_j$ saw $\mathrm{t}_k^{[+wh]}$?]

$\lambda\langle g,w\rangle[\![\text{Which of each other's friends}]\!]^{\vec{g}g,w}$ $(= A)$

$[\![\text{did she}_i \text{ remind Bill that they}_j \text{ saw t}_k^{[+wh]}\ ]\!]^{\vec{g}g^*w^*} =$

$[\![\text{did}]\!]^{g^*w^*}([\![\text{remind}]\!]^{g^*w^*}(g^*(i), \mathit{bill}, \lambda w'[[\![\text{saw}]\!]^{g^*w'}(g^*(j), \vec{g}(k)(\langle g^*, w'\rangle))))]))$

$[\![\text{Which of } \textit{each other}\text{'s friends did she remind Bill that } \textit{they} \text{ saw t }]\!]^{\vec{g}g^*w^*}$

$= \lambda\vec{x}[[\![\text{did she}_i \text{ remind Bill that } \textit{they}_j \text{ saw t}_k^{[+wh]}\ ]\!]^{\vec{g}[k/\vec{x}]g^*w^*}(A)$

$= [[\![\text{did}]\!]^{g^*w^*}([\![\text{remind}]\!]^{g^*w^*}(g^*(i), \mathit{bill}, \lambda w'[[\![\text{saw}]\!]^{g^*w'}(g^*(j), A(\langle g^*, w'\rangle))]))]$

$= [[\![\text{did}]\!]^{g^*w^*}([\![\text{remind}]\!]^{g^*w^*}(g^*(i), \mathit{bill},$
$\lambda w'[[\![\text{saw}]\!]^{g^*w'}(g^*(j), [\![\text{Which of each other's friends}]\!]^{g^*,w'})]))]$

$= [\![\text{did she remind Bill that } \textit{they} \text{ saw Which of } \textit{each other}\text{'s friends}]\!]^{\vec{g}g^*w^*}$

This gives the effect of reconstruction.

Semantics of the interrogative Mood

- The meaning of *did*: $[\![\text{did}]\!]$: a proposion $\rightarrow$ a pair of propositions

  $[\![\text{did}]\!]([\![\phi]\!]) = \{\langle p, \neg p\rangle |\ p \in [\![\phi]\!] \text{ or } p = [\![\phi]\!]\}$

  Assumption: The meaning of an IP is a proposition.

(11) a. Did John laugh?
$\{\lambda w[\![\text{John laughed}]\!]^w, \lambda w[\neg[[\![\text{John laughed}]\!]^w]]\}$

b. Who did John love?
$\{\{\lambda w[[\![\text{loved}]\!]^w(d)(john)], \lambda w[\neg[[\![\text{loved}]\!]^w(d)(john)]]\}|\ d \in D\}$
$= POW(\{\lambda w[[\![\text{loved}]\!]^w(d)(john)]|\ d \in D\})$

Question 2: How is a *wh*-phrase in situ associated with the Mood?

Answer to Question 2: **Alternative non-contextual meanings**

- A *wh*-word generates a set of alternatives, meanings.
- The alternatives also denote non-contextual meanings.

(12) $\text{Who}_j^{[+wh]}$ does John believe every boy_i loves $\text{t}_j^{[+wh]}$?

$\Rightarrow [\![[_M\ \textbf{does}]\ \text{John believe [every boy]}_i \text{ loves } \text{Who}_j^{[+wh]}]\!]^{\vec{g}gw}$

a. $??POW(\{\lambda w'[\text{John believe}^{w'}(\lambda w''[\forall x[\text{boy}^{w''}(x) \rightarrow \text{love}^{w''}(x,y)]])]|$
$y \in Alt([\![\text{who}_j]\!]^{\vec{g}gw})\})$ (main context)

b. $??POW(\{\lambda w'[\text{John believe}^{w'}(\lambda w''[\forall x[\text{boy}^{w''}(x) \rightarrow \text{love}^{w''}(x,y)]])]|$
$y \in Alt([\![\text{who}_j]\!]^{\vec{g}g[i/x]w''})\})$ (local context)

c. $\checkmark POW(\{\lambda w'[\text{John believe}^{w'}(\lambda w''[\forall x[\text{boy}^{w''}(x) \rightarrow \text{love}^{w''}(x,\vec{y}^{g[i/x]w''})]])]|$
$\vec{y} \in Alt([\![\text{who}_j^{[+wh]}]\!]^{\vec{g}[j/\vec{y}]gw})\})$ (non-contextual)

- (12a) leads to the individual reading.

- (12b) is not available because i) x in $g[i/x]$ is free and ii) something introduced in the scope of a universal quantifier or a modal operator is not available in the main context.

(13) a. ??Every man has *a car*. *It* is not fast.

b. ??Amy thinks Bob has *a car*. *It* is not fast.

- (12c) is the only possible reading.

Results:
i) A preposed *wh*-phrase gets the effect of reconstruction and is associated with the Mood by denoting a set of alternative non-contextual meanings.
Question 3: Why is a *wh*-phrase preposed?
ii) A non-contextual meaning is scopeless.
Question 4: How is the scope of a *wh*-phrase determined?

2.3 Why is a *wh*-phrase preposed?

- In languages like Korean and Japanese, no *wh*-movement is necessary.
- In English, if there is more than one *wh*-phrase in a question, only one is preposed.
- The scope of a *wh*-phrase in situ is not subject to syntactic islands, though a *wh*-movement is.

(14) Which linguist will be offended if [we invite *which philosopher*]?

- A preposed *wh*-phrase does not have the scope that a DP normally gets in the surface position. A *wh*-movement is not for the scope of the *wh*-phrase.

(15) A: John is insane. He believes {a hero in a movie, an imaginary creature} attacked him.

B: {Who, What} does John believe attacked him?

A: {Iron Man, A unicorn}.

- The classical analyses of *wh*-questions like Hamblin (1973) and Karttunen (1977) are not quite right.

(16) $\lambda p \exists x[person/animal(x) \wedge p = \hat{}believe(john, \hat{}attack(x, john))]$

Answer to Question 3: In English, a *wh*-movement is to indicate the scope of a question, not the scope of the *wh*-phrase.

2.4 How is the scope of a *wh*-phrase determined?

- A *wh*-phrase is generally taken to be an indefinite.
- An indefinite has three interpretations.

(17) If *a relative of mine* dies, I will inherit a house.

a. If there is *a relative of mine* and he or she dies, I will inherit a house.

b. (I live with three relatives of mine.) If *a relative of mine* dies, ...

c. (I live with a relative mine.) If *the relative of mine* dies, ...

(18) $[\![[\text{a } \alpha]\ \beta]\!] =$

a. $\lambda P \exists x[[\![\alpha]\!](x) \wedge [\![\beta]\!](x)]$ (quantificational use)

b. $\ll \exists X[X \neq \emptyset \wedge \forall x \in X[[\![\alpha]\!](x)]] \gg \exists x' \in X[[\![\beta]\!](x')]$ (split scope)
Only the **descriptive content** is presupposed.

c. $\ll \exists x[[\![\alpha]\!](x)] \gg [\![\beta]\!](x)$ (specific use)
Both the **quantificational force** and the **descriptive content** are presupposed.

- A *wh*-phrase is a specific indefinite, whose scope is determined by presupposition projection of the quantificational force and the descriptive content.

(19) a. If John stops smoking, he will get healthier.
= John has been smoking. If John stops it, he will get healthier.

b. If *a relative of mine* dies, I will inherit a house.
= There is a relative of mine. If she dies, I will inheric a house.

- A *wh*-phrase as a specific indefinite tends to have wide scope, but despite the tendency, it can have narrow or intermediate scope if it is required.

(20) Everyone drove *his car*.
= Everyone who has a car drove it.
≠ There is a car. Every boy drove it.

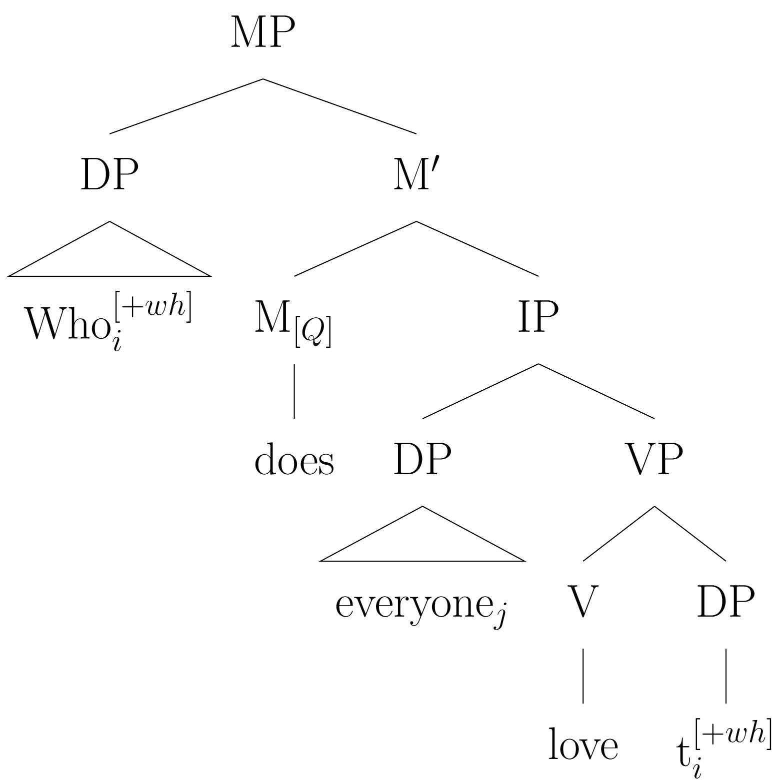
- This explains the functional reading of a *wh*-question with a quantifier.

3 Non-contextual meaning and functional *wh*-questions

3.1 Interpretation of a *wh*-question with the functional reading

- The structure of a *wh*-question with a functional reading:

(21)



Interpretation of a *wh*-phrase as a specific indefinite:
– A *wh*-word generates a set of alternative values of non-contextual meanings

(22) a. $[\![[_{DP}\ which^{[+wh]}\ \alpha]_i^{[+wh]}]\!]^{\vec{g}gw} = \lambda\langle g', w'\rangle [\![[_{DP}\ which^{[+wh]}\ \alpha]_i]\!]^{\vec{g}g'w'}$
$= \{\lambda\langle g', w'\rangle[\ll \pi \gg \vec{x}^{g'[i/y]w'}]\text{: } \vec{x}\in Alt(\vec{g}(i))\}$
$\pi = \exists y[[\![\alpha]\!]^{\vec{g}g'[i/y]w'}(\vec{x}^{g'[i/y]w'})]$, where $\langle g'[i/y], w'\rangle$ is in the domain of $\vec{x}$.

b. π is accommodated in a linguistic context $g''w''$ in the following way:
$\pi \Rightarrow \vec{\pi}$ (abstracted over contexts) $\Rightarrow \vec{\pi}^{g''w''}$ (contextualized)

- Some *wh*-words can be assumed to include some descriptive content.

(23) who = which person;
where = which place;
when = which time
(*which* here is number-neutral)

- Compositional interpretation of a *wh*-question with the functional reading

(24) a. $[\![\mathrm{t}_i^{[+wh]}]\!]^{\vec{g}gw} = \vec{g}(i)^{gw}$

b. $[\![\ \mathrm{t}_j \text{ love } \mathrm{t}_i^{[+wh]}]\!]^{\vec{g}gw} = love^w(g(j), \vec{g}(i)^{gw})$

c. $[\![\,\hat{}\,[\text{everyone}_j\ [\ \mathrm{t}_j \text{ love } \mathrm{t}_i^{[+wh]}]]]\!]^{\vec{g}gw}$
$= \lambda w'[\lambda P \forall x[person^{w'}(x) \to P(x)](\lambda z[\![[\ \mathrm{t}_j \text{ love } \mathrm{t}_i^{[+wh]}]]\!]^{\vec{g}g[j/z]w'})]$
$= \lambda w' \forall x[person^{w'}(x) \to love^{w'}(x, \vec{g}(i)^{g[j/x]w'})]$

d. $[\![\text{does }\hat{}\,[\text{everyone}_j\ [\ \mathrm{t}_j \text{ love } \mathrm{t}_i^{[+wh]}]]]\!]^{\vec{g}gw}$
$= \langle \lambda w' \forall x[person^{w'}(x) \to love^{w'}(x, \vec{g}(i)^{g[j/x]w'})],$
$\lambda w' \neg \forall x[person^{w'}(x) \to love^{w'}(x, \vec{g}(i)^{g[j/x]w'})]\rangle$

e. $[\![\text{Who}_i^{[+wh]}]\!]^{\vec{g}gw} = [\![[\text{Which}^{[+wh]} \text{ person}]_i^{[+wh]}]\!]^{\vec{g}gw}$
$= \{\lambda\langle g', w'\rangle[\ll \pi \gg \vec{x}^{g'[i/y]w'}] : \vec{x} \in Alt(\vec{g}(i))\}$, where
π: $\exists y[person^{w'}(\vec{x}^{g'[i/y]w'})]$

f. $[\![\text{Who}_i^{[+wh]} \text{ does }\hat{}\,[\text{everyone}_j\ [\ \mathrm{t}_j \text{ love } \mathrm{t}_i^{[+wh]}]]]\!]^{\vec{g}gw} =$
$\{\lambda \vec{z}[\![\text{does }\hat{}\,[\text{everyone}_j\ [\ \mathrm{t}_j \text{ love } \mathrm{t}_i^{[+wh]}]]]\!]^{\vec{g}[i/\vec{z}]gw}(\vec{x}) |\ \vec{x} \in [\![\text{Who}_i^{[+wh]}]\!]^{\vec{g}gw}\}$

$= POW(\{\lambda w' \forall x[person^{w'}(x) \to love^{w'}(x, \ll \pi \gg \vec{x}^{g[i/y][j/x]w'})] |\ \vec{x} \in Alt(\vec{g}(i))\}$
$= POW(\{\lambda w' \forall x[person^{w'}(x) \to \ll \exists y[person^{w'}(\vec{x}^{g[i/y][j/x]w'})] \gg$
$love^{w'}(x, \vec{x}^{g[i/y][j/x]w'})] |\ \vec{x} \in Alt(\vec{g}(i))\})$

- The speaker assumes a set of alternatives $\vec{x}$ that look like the one in (25).

(25) $\vec{x} = \begin{bmatrix} g_1 w' & \rightarrow & c_1^{w'} \\ g_2 w' & \rightarrow & c_2^{w'} \\ g_3 w' & \rightarrow & c_3^{w'} \\ ... & ... & ... \end{bmatrix}$, where g_1, g_2, g_3, ... are s.t. $\begin{bmatrix} g_1(j)^{w'} = a_1^{w'} \\ g_2(j)^{w'} = a_2^{w'} \\ g_3(j)^{w'} = a_3^{w'} \\ ... \end{bmatrix}$,
where c_n is in a relation of natural function to a_n.

- What if the presupposition from the *wh*-phrase is accommodated over the universal quantifier, as in (26), with the same alternative non-contextual values of $\vec{x}$?

  Answer: $\vec{x}^{g[i/y][j/x]w'}$ is not defined.

(26) $POW(\{\lambda w' \ll \exists y[person^{w'}(\vec{x}^{g[i/y]w'})] \gg \forall x[person^{w'}(x) \rightarrow$
$love^{w'}(x, \vec{x}^{g[i/y][j/x]w'})]|\ \vec{x} {\in} Alt(\vec{g}(i))\})$

- A case where (1) has an individual reading:

(27) $\vec{x} = \begin{bmatrix} g_1 w & \rightarrow & c_1^{w'} \\ g_2 w & \rightarrow & c_1^{w'} \\ g_3 w & \rightarrow & c_1^{w'} \\ \dots & \dots & \dots \end{bmatrix}$, where g_1, g_2, ... are s.t. $\begin{bmatrix} g_1(j)^{w'} = a_1^{w'} \\ g_2(j)^{w'} = a_2^{w'} \\ g_3(j)^{w'} = a_3^{w'} \\ \dots \end{bmatrix}$

- What if the presupposition from the *wh*-phrase is not projected to the main context?

  Answer: The interpretation itself has no problem, but if a question is asked with the functional reading, non-contextual values like the one in (27) are excluded from the context.

3.2 Problems with a multiply-indexed trace solved

- The meaning of personhood of *who* is reflected in the meaning of $person^{w'}(\vec{x}^{g'[i/y]w'})$ in (24).

(24) f. $[\![\text{Who}_i^{[+wh]} \text{ does } {}^\wedge[\text{everyone}_j \text{ [t}_j \text{ love t}_i^{[+wh]}]]]\!]^{\vec{g}gw} =$
$Pow(\{\lambda w' \forall x[person^{w'}(x) \rightarrow \ll \exists y[person^{w'}(\vec{x}^{g[i/y][j/x]w'})] \gg$
$love^{w'}(x, \vec{x}^{g[i/y][j/x]w'})]|\ \vec{x} \in Alt(\vec{g}(i))\}) \setminus \emptyset$

- A non-contextual value of a trace with one index includes in its domain all possible contextual assignments which assign values to indexes of other DPs.
- We do not have to look at other parts of a sentence in interpreting the trace of a *wh*-phrase.

(28) $\vec{x} = \begin{bmatrix} g_1 w' & \rightarrow & c_1^{w'} \\ g_2 w' & \rightarrow & c_2^{w'} \\ g_3 w' & \rightarrow & c_3^{w'} \\ \ldots & \ldots & \ldots \end{bmatrix}$, where g_1, g_2, ... g_n, ... are s.t.

$g_n(k)^{w'}$ is in a functional relation to $\langle g_n(i)^{w'}, g_n(j)^{w'}\rangle$.

- A short answer can be interpreted with no free variable: The short answer *Their$_j$ mother*, for example, is interpreted as denoting $\lambda\langle g'w'\rangle\iota z[mother^{w'}(z, g'(j))]$ where there is no free variable.

4 Conclusion

A *wh*-phrase introduces a set of alternative non-contextual meanings, which are scopeless, and the scope of a *wh*-phrase, which is a specific indefinite, is determined by the presupposition projection.

Appendix

4.1 Basic semantics for *wh*-questions

- A Model

(29) A Model $M = \langle W, D, F, C\rangle$, where
W is a set of possible worlds,
$D = \bigcup\{D^w |\ w \in W\}$ is a set of individuals,
F is an interpretation function for lexical items, and
C specifies counterpart relations between individuals in different possible worlds.

- Contextual and non-contextual assignments

(30) a. G is a set of functions from a set of natural numbers N into a set of semantic entities of various semantic types.

e.g. $[\![[\mathrm{t}_i]_{DP}]\!]^{\vec{g}gw} = g(i)^w$

b. $\vec{G}$ is a set of functions from N to a set of functions from subsets of $G \times W$ to semantic entities denoted w.r.t. assignment-world pairs in the subsets.
e.g. $[\![[\mathrm{t}_i^{[+wh]}]_{DP}]\!]^{\vec{g}gw} = \vec{g}(i)$, where $\vec{g}(i) = \lambda\langle g'w'\rangle \in \boldsymbol{g} \times \boldsymbol{w}[g'(i)^{w'}]$, where $\boldsymbol{g} \subseteq G$ and $\boldsymbol{w} \subseteq W$.

$$\vec{g}(i) = \begin{bmatrix} g_1 w' & \rightarrow & d_1(= g_1(i)^{w'}) \\ g_2 w' & \rightarrow & d_2(= g_2(i)^{w'}) \\ g_3 w' & \rightarrow & d_3(= g_3(i)^{w'}) \\ \ldots & \ldots & \ldots \end{bmatrix}$$

- A proper name, an intransitive and a transitive verb

(31) a. If α is a proper name, $[\![\alpha]\!]^{\vec{g}gw} = F(\alpha)^w$
e.g. $[\![\mathrm{John}]\!]^{\vec{g}gw} = F(\mathrm{John})^w = jo^w$

b. $[\![[_{IV}\ \alpha\ ]]\!]^{\vec{g}gw} = \lambda x[F(\alpha)^w(x)]$;

c. $[\![[_{TV}\ \alpha\ ]]\!]^{\vec{g}gw} = \lambda y \lambda x[F(\alpha)^w(y)(x)]$

- A VP and an IP

(32) $[\![[_{VP}\ \alpha\ [[_{V}\ \gamma\ ]\ \beta\ ]]]\!]^{\vec{g}gw} = [\![\gamma]\!]^{\vec{g}gw}([\![\beta]\!]^{\vec{g}gw})([\![\alpha]\!]^{\vec{g}gw})$
$[\![[_{IP}\ {}^\wedge\alpha\ ]]\!]^{\vec{g}gw} = \lambda w'[\![\alpha]\!]^{\vec{g}gw'}$

- A quantifier

(33) If Q is a quantificational determiner,
$[\![[[_{DP}\ Q\ \alpha\ ]_i\ \beta\ ]]\!]^{\vec{g}gw} = [[\![Q]\!]^{\vec{g}gw}([\![\alpha]\!]^{\vec{g}gw})(\lambda x^w[[\![\beta]\!]^{\vec{g}g[i/x]w}])]$.

Depending on whether each of the two daughters denotes a set of meanings, the whole expression is interpreted as in (34a), (34b) or (34c). Normal interpretations are defined with A and B and the rest are to deal with non-contextual meanings. In this way, a set of meanings is maintained up to an interrogative M.

(34) $[\![[_{\gamma}\ \alpha\ \beta]]\!]^{\vec{g}gw} =$

a. $\{\ll \pi_1, \pi_2 \gg C : \vec{x} {\in} Alt(\vec{g}(i)), \vec{y} {\in} Alt(\vec{g}(j))\}$ if $[\![\alpha]\!]^{\vec{g}gw} = \{\ll \pi_1 \gg A : \vec{x} {\in} Alt(\vec{g}(i))\}$ & $[\![\beta]\!]^{\vec{g}gw} = \{\ll \pi_2 \gg B : \vec{y} \in Alt(\vec{g}(j))\}$;

b. $\{\ll \pi \gg C : \vec{x} {\in} Alt(\vec{g}(i))\}$ if $[\![\alpha]\!]^{\vec{g}gw} = \{\ll \pi \gg A : \vec{x} {\in} Alt(\vec{g}(i))\}$ & $[\![\beta]\!]^{\vec{g}gw} = B$;

c. C, if $[\![\alpha]\!]^{\vec{g}gw} = A$ and $[\![\beta]\!]^{\vec{g}gw} = B$.
($C = A(B)$ or $B(A)$, depending on the semantic types of α and β.)